

Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,362
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	FAST
Model	llama3.2

■ **AI-ASSISTED REVIEW NOTICE:** This report was generated by an artificial intelligence system using Phi-3 Mini as the language model. It is intended as a supplementary tool to assist human peer reviewers. All findings must be validated by a qualified domain expert before editorial decisions are made. The AI may miss context-specific nuances and should not be used as the sole basis for acceptance or rejection.

Editorial Decision

MINOR REVISION

Weighted Score: 3.7 / 5.0
Confidence: 78%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	4.0	Good	■■■■■
Manuscript Structure	4.0	Good	■■■■■
Methodology & Equations	2.5	Weak	■■■■■
Statistical Analysis	4.2	Good	■■■■■
Results & Evidence	4.0	Good	■■■■■
Discussion & Conclusions	4.2	Good	■■■■■
Figures & Tables	4.0	Good	■■■■■
Scientific Writing	4.0	Good	■■■■■
References & Citations	4.0	Good	■■■■■
Ethics & Transparency	3.5	Good	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 4.0 / 5.0 | Confidence: 100%

StructureReviewer

Manuscript structure and logical coherence

Score: 4.0 / 5.0 | Confidence: 80%

Weaknesses:

- Lack of clear research problem statement in Introduction
- Objectives not explicitly stated

Major Comments:

- Introduction could be improved by clearly stating the research problem and objectives.
- The manuscript lacks a clear conclusion section.

Recommendations:

- Revise Introduction to clearly state the research problem and objectives.
- Add explicit statement of objectives in Methods section.
- Consider adding a Conclusion section that summarizes findings and implications.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 2.5 / 5.0 | Confidence: 70%

Weaknesses:

- The logical flow is not entirely clear, and the sequence of sections could be improved.

Major Comments:

- Consider reordering the sections to improve logical flow.
- Ensure that each section flows naturally from the previous one.

Recommendations:

- Reorder the sections to improve logical flow.
- Provide more context for the methodology and experiments.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: 4.2 / 5.0 | Confidence: 85%

Strengths:

- Bootstrap confidence intervals used
- Holm correction applied correctly
- RMSE, MAE, R² metrics used and interpreted appropriately

Major Comments:

- The choice of Kruskal-Wallis test for comparison may not be appropriate due to the nature of the data (acceleration signals). Consider using parametric tests or non-parametric alternatives like Wilcoxon rank-sum test.

Recommendations:

- Consider providing a power analysis or justification for the sample size used in the study.
- Verify that the assumptions of the Kruskal-Wallis test are met before applying it to the data.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: 4.0 / 5.0 | Confidence: 62%

Strengths:

- 6 figure references were detected in the manuscript.
- 10 table references were detected in the manuscript.
- 52 sequentially numbered equations were detected, suggesting a substantial mathematical formulation.

Weaknesses:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Minor Comments:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Recommendations:

- Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.
-

ResultsReviewer

Results validity and empirical support

Score: 4.0 / 5.0 | Confidence: 80%

Strengths:

- The manuscript presents a clear research objective and framework for the problem.

Weaknesses:

- Some results are not clearly tied to figures or tables, e.g., Pareto analysis and hypervolume/spread indicators.

Major Comments:

- While the manuscript reports various quantitative results, some of them could be more detailed or consistently reported across text, figures, and tables.

Recommendations:

- Provide more detail on Pareto analysis and hypervolume/spread indicators in the Results section.
 - Ensure that all results are clearly tied to their respective figures and tables.
-

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: 4.2 / 5.0 | Confidence: 80%

Weaknesses:

- The discussion could benefit from a more in-depth comparison with prior work, including a critical evaluation of the strengths and limitations of each algorithm.

Major Comments:

- The Discussion section could be improved by explicitly stating the theoretical implications of the EANN framework on structural parameter identification.
- The manuscript would benefit from a clearer explanation of how the computational cost affects the performance of the algorithms.

Recommendations:

- Consider adding more details about the experimental setup and data preprocessing to strengthen the discussion.
- Provide more context on the limitations of classical metaheuristic identification approaches to make the comparison with EANN more meaningful.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **4.0 / 5.0** | Confidence: 80%

Weaknesses:

- type: overused opener; sentence: This paper proposes a novel...
- type: run-on sentence; sentence: Modeling and identification of structural parameters are fundamental to control-oriented civil engineering applications, including structural health monitoring, active, semi-active, and passive vibration control for seismic response mitigation, and state estimation. In numerous studies, structural deformations are effectively represented using lumped mass-spring-damper models [1], [2] which are also widely used in damage detection [3] and observ

Major Comments:

- Consider rephrasing sentences to improve clarity and concision
- Define EANN acronym on first use

Recommendations:

- Revise sentence structure for better flow

ReferencesReviewer

References quality, recency, and relevance

Score: **4.0 / 5.0** | Confidence: 80%

Weaknesses:

- The references section could benefit from more recent references (only 6 since 2021), which may limit the scope of the literature review.
- Some references appear incomplete or truncated, which limits confidence in their accuracy.

Major Comments:

- While the manuscript provides a good balance between classic and recent references, there is a noticeable lack of seminal works that could provide deeper insights into the field.

Recommendations:

- Include more recent references (at least 10) to strengthen the literature review.
- Verify the accuracy of incomplete or truncated references.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **3.5 / 5.0** | Confidence: 60%

Strengths:

- No clear human/animal-subject signal was detected in the parsed manuscript.

Weaknesses:

- Conflict-of-interest declaration was not detected.

Major Comments:

- Conflict-of-interest declaration was not detected.

Recommendations:

- Include a conflict-of-interest declaration.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] Introduction could be improved by clearly stating the research problem and objectives.
- [2] The manuscript lacks a clear conclusion section.
- [3] Consider reordering the sections to improve logical flow.
- [4] Ensure that each section flows naturally from the previous one.
- [5] The choice of Kruskal-Wallis test for comparison may not be appropriate due to the nature of the data (acceleration signals). Consider using parametric tests or non-parametric alternatives like Wilcoxon rank-sum test.
- [6] While the manuscript reports various quantitative results, some of them could be more detailed or consistently reported across text, figures, and tables.
- [7] The Discussion section could be improved by explicitly stating the theoretical implications of the EANN framework on structural parameter identification.
- [8] The manuscript would benefit from a clearer explanation of how the computational cost affects the performance of the algorithms.
- [9] Consider rephrasing sentences to improve clarity and concision
- [10] Define EANN acronym on first use
- [11] While the manuscript provides a good balance between classic and recent references, there is a noticeable lack of seminal works that could provide deeper insights into the field.
- [12] Conflict-of-interest declaration was not detected.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Specific Improvement Recommendations

1. Revise Introduction to clearly state the research problem and objectives.
2. Add explicit statement of objectives in Methods section.
3. Consider adding a Conclusion section that summarizes findings and implications.
4. Reorder the sections to improve logical flow.
5. Provide more context for the methodology and experiments.
6. Consider providing a power analysis or justification for the sample size used in the study.
7. Verify that the assumptions of the Kruskal-Wallis test are met before applying it to the data.
8. Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.
9. Provide more detail on Pareto analysis and hypervolume/spread indicators in the Results section.
10. Ensure that all results are clearly tied to their respective figures and tables.
11. Consider adding more details about the experimental setup and data preprocessing to strengthen the discussion.
12. Provide more context on the limitations of classical metaheuristic identification approaches to make the comparison with EANN more meaningful.

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